Mano Engineering LLC
2512 Northland Drive, Mendota Heights, MN 55120

出貨檢查記錄表 Inspection Report

產品編號 MK's P/N	Z0064-004	料號 Customer P/N	F008
生產批號 Lot No.	P-02601052	品名 Part Name	SPRING, COMPRESSION

檢驗日期 Date	February 23, 2026			
批量 Lot Size	1,400			
抽樣標準 Sampling Criteria	允收 Accept	1	拒收 Reject	2
抽樣數 Sample Size	10			

檢驗量具代號 INSPECTION GAUGES

1.游標卡尺Vernier Caliper 2.分厘卡Micrometer 3.捲尺Measuring Tape
4.拉壓力測試機Spring Force Tester 5.扭力荷重機Torque Tester 6.塞規/圓棒Pin Gauge
7.治具Fixture 8.高度規Height Gauge 9.硬度計Hardness Tester 10. 投影機Projector
11.目視Visual 12.其他Others

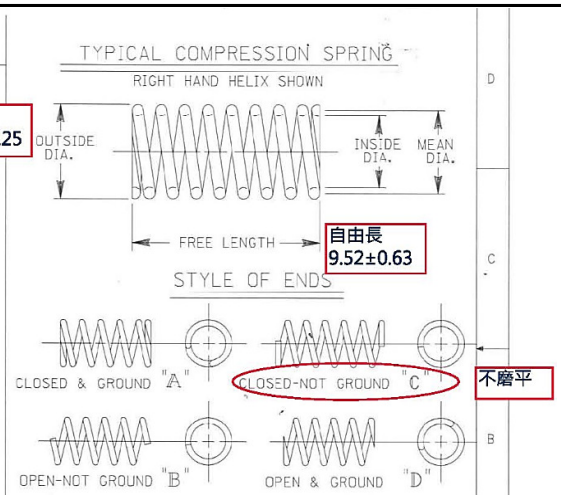
DO NOT SCALE DRAWINGS

SPECIFICATIONS

1. FREE LENGTH .400/.350
2. CONTROLLING DIA.
   A. OUTSIDE DIA. .260/.240 WORKS INSIDE
   B. INSIDE DIA. WORKS OVER
   C. MEAN DIA.
3. WIRE SIZE .016 Ø
4. TOTAL COILS 7.75 (REF.) ACTIVE COILS 3.75 (REF.)
5. SPRING RATE (LBS. PER IN.) 1.75 ± .17 LBS.
6. LOAD AT IN. LBS.
   LOAD AT IN. LBS.
7. STYLE OF ENDS "C" CLOSED & NOT GROUND - SEE OTHER
8. SOLID HEIGHT
9. DIRECTION OF HELIX (COILS) L.H.
10. MATERIAL MUSIC WIRE PER ASTM A228
11. FINISH COATING OIL
12. OTHER 2 CLOSED COILS - EACH END

外徑 6.35±0.25

線徑0.4mm



項目 Item	檢驗項目 Description	規格 Spec. & Tol.	量具 Gauge Code	檢驗結果 Data 1	2	3	4	5	6	7	8	9	10
				Unit: millimeter (mm)									
1	Material	SWPB	12	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
2	Wire Dia.	0.4	2	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
3	Outside Dia	6.35±0.25	1	6.31	6.32	6.33	6.32	6.31	6.32	6.33	6.33	6.31	6.32
4	Free Length	9.52±0.63	1	9.55	9.50	9.52	9.55	9.53	9.55	9.53	9.52	9.53	9.50
5	Total Coils	(7.75T)	11	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5
6	Wind	Left Hand	11	L	L	L	L	L	L	L	L	L	L
7	Finish	Oil	11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
8	Ground	Ends Closed Not Ground	11	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
9	Rate	1.75±0.17LBS/IN (31.24gf±3.04/mm)	4										
10													
11													
12													
13													

主管 Supervisor	檢驗 Inspector	判定Judgement
品管部 2026.02.23 黃嫻穎	品管部 2026.02.23 陳怡伶	合格 OK
		不合格 NG

No. 1			No. 6		
1	8.00 mm	45.0 gf	1	8.00 mm	50.7 gf
2	6.00 mm	107.8 gf	2	6.00 mm	113.5 gf
K		31.41 gf/mm	K		31.41 gf/mm
No. 2			No. 7		
1	8.00 mm	47.5 gf	1	8.00 mm	47.0 gf
2	6.00 mm	110.2 gf	2	6.00 mm	109.5 gf
K		31.36 gf/mm	K		31.26 gf/mm
No. 3			No. 8		
1	8.00 mm	47.5 gf	1	8.00 mm	49.6 gf
2	6.00 mm	109.8 gf	2	6.00 mm	112.7 gf
K		31.15 gf/mm	K		31.56 gf/mm
No. 4			No. 9		
1	8.00 mm	47.7 gf	1	8.00 mm	50.1 gf
2	6.00 mm	110.4 gf	2	6.00 mm	113.0 gf
K		31.36 gf/mm	K		31.46 gf/mm
No. 5			No. 10		
1	8.00 mm	48.1 gf	1	8.00 mm	47.3 gf
2	6.00 mm	111.0 gf	2	6.00 mm	110.3 gf
K		31.41 gf/mm	K		31.51 gf/mm

Mano Engineering LLC

Address: 2512 Northland Drive, Mendota Heights, MN 55120

Sheet No. 2024T-515-01

Date APR.26.2024

TEST CERTIFICATE

L/C No.

Order No. SW-H-312

Commodity STAINLESS STEEL WIRE NI-CO AISI 304

Specification JIS G 4314 WPB

Grade of Wire Quality 2050 - 2300 N/㎟

Nominal Dia. of Wire 0.400 mm

Order Quantity 20 KGS X 20 C/L

Net Weight 400.00 KGS

Gross Weight

TEST RESULT

Sample No.	Dia of Wire (mm)	Dia Ovality (mm)	Breaking Load (N)	Tensile Strength of Wire (N/㎟)	Elongation (%) ( )	Reduction of area (%)	Number of Torsion (Dia x)	Wrap Test (Dia x)	Bend Test (R= mm)
001	0.3990	0.0020		2,228.0					
002	0.3990	0.0020		2,216.0					
003	0.3990	0.0020		2,243.0					
004	0.3990	0.0020		2,201.0					
005	0.3990	0.0020		2,220.0					
006	0.3990	0.0020		2,191.0					
007	0.3990	0.0020		2,208.0					

REMARK

CHEMICAL ANALYSIS OF WIRE ROD

Heat No.	C	Si	Mn	P	S	Ni	Cr	Cu	N
S56860	0.073	0.375	1.245	0.034	0.001	8.08	18.42		

Lee.M.J

Chief of Quality Control Dept.

MHS-B102-34

(203 x 280)